

HCSM-12 (Second Revision)

The Hodge HCSM Interface: Entropic Flatness The Maximally Mixed State on the 56-Dimensional Hodge Sector and the Processor Channel The Hodge Complex Standard Model

Stanley Preschutti*

September 2026

Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04), and on the full 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$, the processor T selects a fermion sector of dimension $|F| = 10$ and a boson sector of dimension $|B_{\text{bos}}| = 4$ (HCSM-13). This paper establishes the entropic properties of the Hodge HCSM interface. We prove: (i) the natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, obtained as the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection of HCSM-10; its von Neumann entropy is $S_{\text{in}} = \log 56$; (ii) the processor T extends to a rank-40 orthogonal projection $T_{56} = \text{extend}(T)$ on V_{56} (HCSM-16, Definition 1), and the associated processor channel is a completely positive trace-preserving map whose output on the maximally mixed input is the maximally mixed state on the 40-dimensional image of T_{56} , with entropy $S_{\text{out}} = \log 40$; (iii) the entropy reduction is $\Delta S = \log 56 - \log 40 = \log(7/5) \approx 0.33647$ nats; (iv) the reduction equals the exact deterministic Landauer bound for the processor's information-processing step, conditional on the HCSM measurement-irreversibility principle. All claims are HCSM-native and verified at machine precision.

Contents

1	Introduction	2
1.1	The Hodge–HCSM interface	2
1.2	The one line summary	3
1.3	Main results	3
1.4	Cross references to the HCSM paper stack	3
1.5	Organization	3
2	Setup	4
2.1	The 56-dimensional Hodge sector	4
2.2	The HCSM processor	4
2.3	Extension of the processor to V_{56}	4

*Entropia Research Institute / Information Physics Institute.
ORCID: 0009-0004-5445-1744. Email: scstamp@yahoo.com.

3	Entropic flatness of the Hodge output	5
3.1	The full Hodge complex and its maximally mixed state	5
3.2	Mirror cancellation and the projection onto V_{56}	5
4	The processor channel	6
4.1	The CPTP map	6
4.2	The output state	6
5	The entropy reduction	6
6	Landauer saturation	7
7	Interpretation	7
7.1	The Hodge–HCSM interface as an information-theoretic transition	7
7.2	The interface preserves a specific structure	8
7.3	Relation to the Landauer bound	8
7.4	What the interface does not claim	8
8	Inputs and derived	8
8.1	Derived	8
8.2	Inputs	9
8.3	Disclosed inputs	9
8.4	What is not claimed	9
9	Reproducibility	9
10	Conclusion	10
A	Web Crawler Metadata	11
A.1	Dublin Core	11
A.2	BibTeX	12

1 Introduction

1.1 The Hodge–HCSM interface

The Hodge Complex Standard Model (HCSM) has two structural levels. The first is the *Hodge complex*: the exterior algebra of differential forms $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ on the periodic 8×8 torus (HCSM-02 [3]), with the Hodge–Dirac operator $D = d + d^*$ and the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian. The second is the *HCSM processor*: the rank-10 orthogonal projection T on the 14-mode multiplet at $\lambda = 4$, which separates the fermion sector F ($|F| = 10$) from the boson sector B_{bos} ($|B_{\text{bos}}| = 4$) (HCSM-13 [7]).

The *Hodge HCSM interface* is the passage from the abstract Hodge complex to the processed fermion/boson structure. This paper establishes the information-theoretic properties of that interface: the entropy of the state the Hodge complex delivers, and the entropy of the state the processor produces.

1.2 The one line summary

The natural state on the 56-dimensional Hodge sector is the maximally mixed state, with entropy $\log 56$ nats. The processor T maps this to the maximally mixed state on its 40-dimensional image, with entropy $\log 40$ nats. The entropy reduction is exactly $\log(7/5) \approx 0.33647$ nats, saturating the deterministic Landauer bound.

1.3 Main results

- (1) **Entropic flatness** (Theorem 3.1). The natural state on the 56-dimensional Hodge sector at $\lambda = 4$ is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, obtained as the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection of HCSM-10. Its von Neumann entropy is $S_{\text{in}} = \log 56$ nats.
- (2) **Processor channel** (Theorem 4.2). The processor T extends to a rank-40 orthogonal projection $T_{56} = \text{extend}(T)$ on V_{56} (HCSM-16, Definition 1), and the associated processor channel is a completely positive trace-preserving map. Applied to ρ_{in} , it yields the maximally mixed state on $\text{im}(T_{56})$ of dimension 40, with entropy $S_{\text{out}} = \log 40$ nats.
- (3) **Entropy reduction** (Corollary 5.1). $\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log(7/5) \approx 0.33647$ nats.
- (4) **Landauer saturation** (Theorem 6.1). The reduction $\Delta S = \log(7/5)$ equals the deterministic Landauer bound for the processor's information-processing step, conditional on the HCSM measurement-irreversibility principle (Principle 6).
- (5) **Framework-constant characterization** (Corollary 5.2). The reduction has the clean HCSM-native form $\Delta S = \log(\dim V_{56} / \text{rank}(T_{56})) = \log(56/40) = \log(7/5)$.

1.4 Cross references to the HCSM paper stack

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the framework constants.
- HCSM-02 [3]: the Hodge complex; the 56-dimensional sector.
- HCSM-04 [4]: the 14-mode multiplet.
- HCSM-05 [5]: D_4 representation theory of the 14-mode multiplet.
- HCSM-10 [6]: the mirror-cancellation theorem; residual $7/32$.
- HCSM-13 [7]: the processor T ; the F/B_{bos} split.
- HCSM-16 [8]: the loop operator on the full 56-dimensional Hodge sector; the block-diagonal extension map.

1.5 Organization

Section 2 states the Hodge sector and the processor. Section 3 proves the entropic-flatness theorem. Section 4 proves the processor-channel theorem. Section 5 derives the entropy reduction. Section 6 establishes the Landauer saturation as a conditional theorem. Section 7 gives the information-theoretic interpretation. Section 8 summarizes inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The 56-dimensional Hodge sector

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01 [2]). The Hodge complex at the self-paired eigenvalue $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (1)$$

of total dimension $14 + 28 + 14 = 56$ (HCSM-02 [3]). Its D_4 isotypic decomposition is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E, \quad (2)$$

with dimension check $9 + 5 + 7 + 7 + 28 = 56$ (HCSM-02 [3], HCSM-05 [5]).

2.2 The HCSM processor

On the 14-mode multiplet at $\lambda = 4$, the processor T is the rank-10 orthogonal projection (HCSM-13 [7])

$$T = P_{\text{mesh}} + P_\tau = I_{14} - P_C, \quad (3)$$

with image the fermion sector F and kernel the boson sector B_{bos} :

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad (4)$$

$$B_{\text{bos}} = \{\text{dark}, W, Z, H\}, \quad |B_{\text{bos}}| = 4. \quad (5)$$

The classes A , B_{wind} , C are the winding classes of HCSM-14; the boson sector is the winding class C .

2.3 Extension of the processor to V_{56}

By HCSM-16 (Definition 1, Lemma 2), the block-diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$,

$$\text{extend}(G) = \begin{pmatrix} G & 0 & 0 \\ 0 & G \otimes \mathbb{I}_2 & 0 \\ 0 & 0 & G \end{pmatrix}, \quad (6)$$

in the block ordering $\mathbb{C}^{14} \oplus (\mathbb{C}^{14} \otimes \mathbb{C}^2) \oplus \mathbb{C}^{14}$ is a Lie algebra homomorphism. We denote by

$$T_{56} = \text{extend}(T) = T \oplus (T \otimes \mathbb{I}_2) \oplus T \quad (7)$$

the 56-dimensional processor.

Lemma 2.1 (Processor rank on V_{56}). *The operator T_{56} is an orthogonal projection with*

$$\text{rank}(T_{56}) = 10 + 20 + 10 = 40, \quad \dim \ker(T_{56}) = 16. \quad (8)$$

Proof. T is a rank-10 projection on $\mathbb{C}^{14}$ with $T^2 = T$ and $T^\dagger = T$. The block-diagonal structure (7) inherits idempotence and Hermiticity. The rank is $\text{rank}(T) + \text{rank}(T \otimes \mathbb{I}_2) + \text{rank}(T) = 10 + 20 + 10 = 40$. The kernel has dimension $56 - 40 = 16$. $\square$

Lemma 2.2 (Fermion/boson structural decomposition). *The processor T_{56} distinguishes a fermion sector of dimension $|F| = 10$ and a boson sector of dimension $|B_{\text{bos}}| = 4$ per form degree; on V_{56} the processor image has dimension*

$$\text{rank}(T_{56}) = |F| \cdot (1 + 2 + 1) = 10 \cdot 4 = 40. \quad (9)$$

Proof. By Lemma 2.1, $\text{rank}(T_{56}) = 40$. The factor $1 + 2 + 1$ counts the contributions to Ω^0 , Ω^1 , and Ω^2 respectively. $\square$

3 Entropic flatness of the Hodge output

3.1 The full Hodge complex and its maximally mixed state

The full Hodge complex on Λ has dimension $\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2) = 64 + 128 + 64 = 256$ (HCSM-02 [3]). We take as the natural state on the full Hodge complex the maximally mixed state

$$\rho_{\text{full}} = \frac{\mathbb{I}_{256}}{256}, \quad (10)$$

the unique unit-trace state invariant under the full unitary group $U(256)$ preserving the Hodge inner product. This is a maximum-entropy principle, not a D_4 -uniqueness claim: on a reducible representation, the D_4 commutant is strictly larger than the scalars, so D_4 invariance alone does not single out $\mathbb{I}/\dim$.

3.2 Mirror cancellation and the projection onto V_{56}

By HCSM-10 [6], the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the 5-point Laplacian spectrum pairs 50 of the 64 scalar modes into cancelling partners; only the 14 self-paired modes at $\lambda = 4$ survive. On the full Hodge complex this is the orthogonal projection $P_{\text{mir}} \equiv P_{\lambda=4}$ onto the 56-dimensional self-paired eigenspace V_{56} (1).

The mirror-cancellation projection defines a completely positive trace-preserving (CPTP) map

$$\Phi_{\text{mir}}(\rho) = P_{\text{mir}} \rho P_{\text{mir}} + \text{Tr}((\mathbb{I}_{256} - P_{\text{mir}})\rho) |0\rangle\langle 0|_{\text{bath}}, \quad (11)$$

where $|0\rangle_{\text{bath}}$ is a fixed reference state on the discarded 200-dimensional bath.

Theorem 3.1 (Entropic flatness). *The natural state on the 56-dimensional Hodge sector V_{56} is the maximally mixed state*

$$\rho_{\text{in}} = \Phi_{\text{mir}}(\rho_{\text{full}}) = \frac{\mathbb{I}_{56}}{56}, \quad (12)$$

the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection. Its von Neumann entropy is

$$S_{\text{in}} = -\text{Tr}(\rho_{\text{in}} \log \rho_{\text{in}}) = \log 56 \approx 4.0253516907 \text{ nats}. \quad (13)$$

Proof. Since P_{mir} is a projection and $\rho_{\text{full}} = \mathbb{I}_{256}/256$,

$$P_{\text{mir}} \rho_{\text{full}} P_{\text{mir}} = \frac{P_{\text{mir}}}{256} = \frac{\mathbb{I}_{56} \oplus 0_{200}}{256}. \quad (14)$$

Taking the trace and normalizing yields $\rho_{\text{in}} = \mathbb{I}_{56}/56$. The entropy follows:

$$S_{\text{in}} = -\sum_{i=1}^{56} \frac{1}{56} \log \frac{1}{56} = \log 56. \quad (15)$$

□

Remark 3.2 (Why maximally mixed). The state $\rho_{\text{in}} = \mathbb{I}_{56}/56$ is the unique unit-trace state invariant under the full unitary group $U(56)$ acting on V_{56} . The derivation via the mirror-cancellation projection is HCSM-native: the projection onto the self-paired eigenspace is the structural content of the mirror-cancellation theorem (HCSM-10).

Remark 3.3 (Entropic flatness). The term *entropic flatness* refers to the fact that ρ_{in} is uniform over the 56 dimensions of V_{56} : every dimension carries equal weight $1/56$. This is the maximum-entropy state on V_{56} .

4 The processor channel

4.1 The CPTP map

Definition 4.1 (Processor channel). *The processor channel is the CPTP map*

$$\Phi_T(\rho) = T_{56} \rho T_{56} + \text{Tr}((\mathbb{I}_{56} - T_{56})\rho) |0\rangle\langle 0|_{\text{bos}}, \quad (16)$$

with $T_{56} = \text{extend}(T)$ from (7) and $|0\rangle_{\text{bos}}$ a fixed reference state on the 16-dimensional bosonic kernel.

The map Φ_T is completely positive because it is a sum of two completely positive maps ($\rho \mapsto T_{56}\rho T_{56}$ and the trace-scaling channel onto $|0\rangle\langle 0|_{\text{bos}}$), and it is trace-preserving because $\text{Tr}(T_{56}\rho T_{56}) + \text{Tr}((\mathbb{I} - T_{56})\rho) = \text{Tr}(\rho)$.

4.2 The output state

Theorem 4.2 (Processor channel output). *The processor channel Φ_T maps the maximally mixed input $\rho_{\text{in}} = \mathbb{I}_{56}/56$ to the maximally mixed state on the 40-dimensional image of T_{56} :*

$$\rho_{\text{out}} \propto \Phi_T(\rho_{\text{in}}) = \frac{T_{56}}{56} = \frac{\mathbb{I}_{\text{im}(T_{56})} \oplus 0_{16}}{56}. \quad (17)$$

Restricted to the image of T_{56} and renormalized, $\rho_{\text{out}}|_{\text{im}(T_{56})} = \mathbb{I}_{40}/40$, with von Neumann entropy

$$S_{\text{out}} = \log 40 \approx 3.6888794541 \text{ nats}. \quad (18)$$

Proof. Since T_{56} is a projection ($T_{56}^2 = T_{56}$) and $\rho_{\text{in}} = \mathbb{I}_{56}/56$,

$$T_{56} \rho_{\text{in}} T_{56} = \frac{T_{56}}{56}. \quad (19)$$

The trace of this term is $\text{rank}(T_{56})/56 = 40/56 = 5/7$. The complementary term $\text{Tr}((\mathbb{I} - T_{56})\rho_{\text{in}})|0\rangle\langle 0|_{\text{bos}}$ contributes a pure state on the 16-dimensional kernel with weight $16/56 = 2/7$. The marginal on $\text{im}(T_{56})$ is therefore $\mathbb{I}_{40}/40$, with entropy $\log 40$. $\square$

Remark 4.3 (Image-of- T_{56} language). The output state is naturally a state on the image of T_{56} , of dimension 40. The state is not, strictly speaking, a state on the product space $F \otimes B_{\text{bos}}$; the language of “fermion–boson product space” of the first revision should be replaced by the language of the image of T_{56} .

5 The entropy reduction

Corollary 5.1 (Entropy reduction). *The entropy reduction effected by the processor is*

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log\left(\frac{56}{40}\right) = \log\left(\frac{7}{5}\right) \approx 0.3364722366 \text{ nats}. \quad (20)$$

Proof. By Theorem 3.1, $S_{\text{in}} = \log 56$. By Theorem 4.2, $S_{\text{out}} = \log 40$. Subtract. $\square$

Corollary 5.2 (Framework-constant form). *The entropy reduction has the clean HCSM-native form*

$$\Delta S = \log\left(\frac{\dim V_{56}}{\text{rank}(T_{56})}\right) = \log\left(\frac{56}{40}\right) = \log\left(\frac{7}{5}\right). \quad (21)$$

The numerator $\dim V_{56} = 56$ is the dimension of the Hodge sector; the denominator $\text{rank}(T_{56}) = 40$ is the rank of the processor. Both are framework constants at $d = 4$.

Proof. Immediate from (1), Lemma 2.1, and Corollary 5.1. $\square$

Remark 5.3 (Ratio in bits). The entropy reduction in bits is

$$\Delta S / \log 2 = \log_2(7/5) \approx 0.4854268272 \text{ bits}. \quad (22)$$

6 Landauer saturation

The entropy reduction $\Delta S = \log(7/5)$ of Corollary 5.1 is a structural identity of the HCSM state space. Whether it equals the *thermodynamic* Landauer bound is a physical statement that requires an additional principle: the identification of the entropy reduction with heat dissipated into the DME bath at the rate $1/T_{\text{DME}}$.

[Measurement irreversibility] The processor channel Φ_T is a non-unitary projection. The entropy reduction $\Delta S = \log(7/5)$ is transferred to the DME bath at the rate $1/T_{\text{DME}}$, where $T_{\text{DME}} = 1/(2\gamma)$ is the unique dissipative time scale of the HCSM five-layer loop (HCSM-21 [9], HCSM-54 [10]).

Theorem 6.1 (Landauer saturation, conditional). *Conditional on Principle 6, the entropy reduction $\Delta S = \log(7/5)$ equals the exact deterministic Landauer bound for the processor's information-processing step. The processor erases exactly $\log(7/5)$ nats of entropy per pass, at the exact deterministic limit. The loop-pass duration is $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$.*

Proof. The processor channel Φ_T is a deterministic rank-40 projection with well-defined image and kernel; no randomness enters its action. By Principle 6, the entropy reduction is transferred to the DME bath at the rate $1/T_{\text{DME}}$. The deterministic Landauer bound is therefore saturated exactly, and the loop-pass duration equals the dissipative time T_{DME} . $\square$

Remark 6.2 (Determinism of the processor). The processor T_{56} is a fixed linear map with a well-defined image (the fermion sector on each form degree) and a well-defined kernel (the boson sector on each form degree). No randomness enters its action. The Landauer bound is therefore saturated exactly, conditional on Principle 6.

Remark 6.3 (Status of the conditional). The Landauer saturation is a *conditional* theorem: the structural identity $\Delta S = \log(7/5)$ is unconditional, but the identification with the thermodynamic Landauer bound requires Principle 6. This principle is HCSM-native (it is the content of HCSM-21, Theorem 12.1) but is not derived from the substrate alone.

7 Interpretation

7.1 The Hodge–HCSM interface as an information-theoretic transition

The Hodge complex provides the substrate's raw information content: a flat (maximally mixed) state on the 56-dimensional Hodge sector, of entropy $\log 56$ nats. The processor T_{56} reads out the fermion/boson structure, reducing the entropy to $\log 40$ nats.

The interface therefore performs an information-theoretic transition from a flat state on 56 dimensions to a flat state on 40 dimensions. The reduction is exactly $\log(7/5) \approx 0.33647$ nats. This is the information the processor extracts, and it is the amount of entropy the substrate’s raw state must release for the fermion/boson structure to be observable.

7.2 The interface preserves a specific structure

The reduction preserves the fermion/boson factorization. The state before processing is flat on all 56 dimensions; the state after processing is flat on the 40-dimensional image of T_{56} . The processor’s information content is the fermion/boson label structure; the finer structure of the Hodge complex (the individual D_4 multiplicities, the class partition) is not directly observable at the level of the interface.

7.3 Relation to the Landauer bound

The processor is a Maxwell demon in the exact deterministic limit, conditional on Principle 6. The Landauer bound is the thermodynamic constraint on the amount of entropy that can be erased per bit of information processing. The processor saturates this bound exactly: $\Delta S = \log(7/5) = \log(\dim V_{56} / \text{rank}(T_{56}))$.

The fact that the reduction saturates the bound, rather than being strictly larger, is a statement about the optimality of the processor: it erases exactly the entropy needed to convert the raw Hodge state into the fermion/boson output state, and no more.

7.4 What the interface does not claim

The interface does not claim to derive any particle properties (masses, charges, mixings) from the entropic reduction. Those are the subject of downstream papers (HCSM-23, HCSM-30, etc.). The interface establishes only the information-theoretic transition: what state the Hodge complex delivers, what state the processor produces, and how much entropy is reduced.

The interface also does not claim that the entropy reduction is a fundamental physical process. It is a structural identity: the difference between two specific entropies computed from the HCSM state space.

8 Inputs and derived

8.1 Derived

1. The entropic flatness of the Hodge output: $S_{\text{in}} = \log 56$ (Theorem 3.1).
2. The processor channel: $S_{\text{out}} = \log 40$ (Theorem 4.2).
3. The entropy reduction: $\Delta S = \log(7/5)$ (Corollary 5.1).
4. The framework-constant form of the reduction (Corollary 5.2).
5. The Landauer saturation, conditional on Principle 6 (Theorem 6.1).

8.2 Inputs

1. The 56-dimensional Hodge sector V_{56} at $\lambda = 4$ (HCSM-02 [3]).
2. The fermion and boson sector dimensions $|F| = 10$, $|B_{\text{bos}}| = 4$ (HCSM-13 [7]).
3. The framework constants $d = 4$, $N = 64$, $L = 8$ (HCSM-01 [2]).
4. The block-diagonal extension map `extend` (HCSM-16 [8]).
5. The mirror-cancellation projection onto the self-paired eigenspace (HCSM-10 [6]).
6. The standard density-matrix formalism: von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$, maximally mixed state $\rho = \mathbb{I}/d$, and the Landauer bound ([11], [12]).

8.3 Disclosed inputs

1. The maximally mixed state on the full Hodge complex $\rho_{\text{full}} = \mathbb{I}_{256}/256$ (a maximum-entropy principle).
2. The measurement-irreversibility principle (Principle 6) for the Landauer interpretation of the entropy reduction.

8.4 What is not claimed

We do not claim that the entropy reduction $\log(7/5)$ has any direct observational consequence. It is a structural identity of the HCSM state space. We also do not claim that a physical process produces the flat state on the Hodge sector: the flat state is the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection, and its flatness is a property of the Hodge–HCSM interface.

9 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-12_verification.py`. It verifies:

1. The dimension of the 56-dimensional Hodge sector.
2. The fermion and boson sector dimensions $|F| = 10$, $|B_{\text{bos}}| = 4$.
3. The processor extension $T_{56} = \text{extend}(T)$ and $\text{rank}(T_{56}) = 40$.
4. The entropy $S_{\text{in}} = \log 56 \approx 4.0253516907$.
5. The entropy $S_{\text{out}} = \log 40 \approx 3.6888794541$.
6. The reduction $\Delta S = \log(7/5) \approx 0.3364722366$.
7. The reduction in bits $\Delta S / \log 2 \approx 0.4854268272$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

This paper has established the entropic properties of the Hodge–HCSM interface of the Hodge Complex Standard Model.

The natural state on the 56-dimensional Hodge sector at $\lambda = 4$ is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, obtained as the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection of HCSM-10. Its von Neumann entropy is $S_{\text{in}} = \log 56 \approx 4.02535$ nats.

The processor T extends to a rank-40 orthogonal projection $T_{56} = \text{extend}(T)$ on V_{56} (HCSM-16), and the associated processor channel Φ_T maps ρ_{in} to the maximally mixed state on $\text{im}(T_{56})$ of dimension 40, with entropy $S_{\text{out}} = \log 40 \approx 3.68888$ nats. The entropy reduction is

$$\Delta S = \log 56 - \log 40 = \log(7/5) \approx 0.33647 \text{ nats.} \quad (23)$$

The reduction saturates the deterministic Landauer bound, conditional on the HCSM measurement-irreversibility principle (Principle 6).

The Hodge–HCSM interface is therefore an information-theoretic transition from a flat state on 56 dimensions to a flat state on 40 dimensions, with entropy reduction $\log(7/5)$. The transition is structural, exact, and HCSM-native. It is the natural starting point for downstream information-theoretic analyses of the processor and its outputs.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The $\log(7/5)$ is what the machine cannot cancel.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: D_4 Representation Theory of the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-10: The Mirror Cancellation Theorem*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-16: Loop Operator on the Full 56-Dimensional Hodge Sector*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-21: The Time Theorem*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-54: Vacuum Stability and the Stability Coefficient*, HCSM White Paper Series (2026).
- [11] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM J. Res. Dev. **5**, 183 (1961).

- [12] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2000).

A Web Crawler Metadata

A.1 Dublin Core

DC.title HCSM-12 (Second Revision): The Hodge–HCSM Interface: Entropic Flatness

DC.creator Preschutti, Stanley

DC.creator.orcid
0009-0004-5445-1744

DC.date.issued
2026-09

DC.date.modified
2026-09

DC.type Text / Preprint / White Paper

DC.format application/pdf; text/x-latex

DC.language
en

DC.publisher
Entropia Research Institute

DC.rights All rights reserved

DC.subject Hodge Complex Standard Model; HCSM; entropic flatness; Hodge–HCSM interface; maximally mixed state; von Neumann entropy; Landauer bound; fermion/boson split; 56-dimensional Hodge sector; processor channel; CPTP map

DC.description
The Hodge Complex Standard Model’s Hodge–HCSM interface delivers a maximally mixed state on the 56-dimensional Hodge sector at $\lambda = 4$, with von Neumann entropy $S_{\text{in}} = \log 56$. The state is the image of the maximally mixed state on the full Hodge complex under the mirror-cancellation projection of HCSM-10. The processor T extends to a rank-40 orthogonal projection T_{56} on V_{56} (HCSM-16), and the associated CPTP channel maps ρ_{in} to the maximally mixed state on $\text{im}(T_{56})$, with entropy $S_{\text{out}} = \log 40$. The entropy reduction is $\Delta S = \log 56 - \log 40 = \log(7/5) \approx 0.33647$ nats, saturating the deterministic Landauer bound conditional on the HCSM measurement-irreversibility principle.

DC.identifier
HCSM-12-2026-09-Rev2

A.2 BibTeX

```
@article{PreschuttiHCSM12Rev2,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-12} (Second Revision): The Hodge-{HCSM}  
    Interface: Entropic Flatness},  
  journal = {HCSM White Paper Series},  
  year = {2026},  
  month = {September},  
  note = {ORCID: 0009-0004-5445-1744},  
  keywords = {Hodge Complex Standard Model, HCSM, entropic  
    flatness, Hodge-HCSM interface, maximally mixed  
    state, von Neumann entropy, Landauer bound,  
    fermion/boson split, processor channel, CPTP map},  
  abstract = {The Hodge-HCSM interface delivers a maximally mixed  
    state on the 56-dimensional Hodge sector at  
     $\lambda = 4$ , with entropy  $S_{\text{in}} = \log 56$ . The state  
    is the image of the maximally mixed state on the  
    full Hodge complex under the mirror-cancellation  
    projection of HCSM-10. The processor  $T$  extends to  
    a rank-40 orthogonal projection  $T_{56}$  on  $V_{56}$   
    (HCSM-16), and the associated CPTP channel maps  
     $\rho_{\text{in}}$  to the maximally mixed state on  $\text{im}(T_{56})$ ,  
    with entropy  $S_{\text{out}} = \log 40$ . The entropy reduction  
    is  $\Delta S = \log 56 - \log 40 = \log(7/5)$  approx  
    0.33647 nats, saturating the deterministic Landauer  
    bound conditional on the HCSM  
    measurement-irreversibility principle.}  
}
```